

Session Objectives

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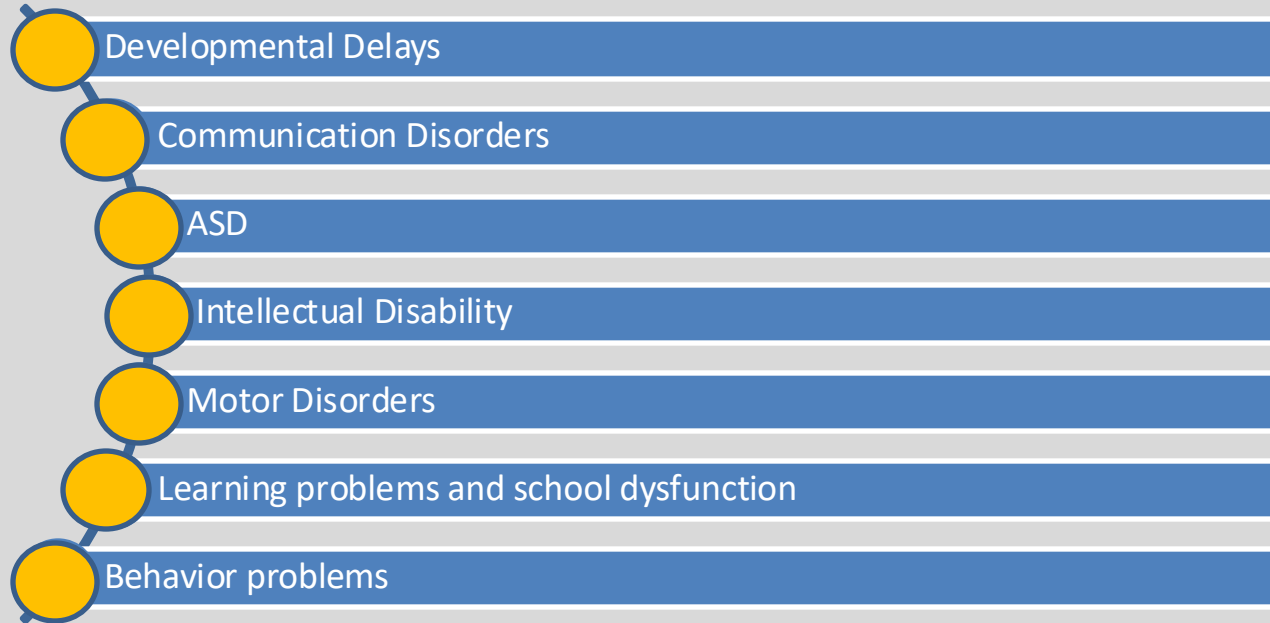
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- Compare and contrast the diagnostic features for speech delay, the communication disorders, and autism spectrum disorder, providing a differential diagnosis, workup and management for each.
- Define intellectual disability, identify potential causes, and describe the workup for a child suspected of having intellectual disability.
- Compare and contrast chronic tic disorder and Tourette syndrome.
- Explain the definition of learning disorder, describe the important elements of the history and workup for a child presenting with a suspected learning disorder and outline the differential diagnosis.
- Compare and contrast ODD and CD and outline the elements of the history that are important for a child presenting with a behavior problem.

OUTLINE

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Patient 1

- 24 month old male at your office for a well visit
- Mother's concerns:
 - Child not saying as many words as sibling
 - Child has frequent tantrums
- Birth Hx: prematurity at 34 weeks of gestation
- FH: learning problems, ADHD
- Exam: normal growth parameters and physical exam
- Observations during encounter: poor eye contact at times, does not respond to name, only says "no", plays with toy car

Patient 1: Differential Diagnosis

- Expressive speech delay
- Communication disorder
- Hearing loss
- Autism spectrum disorder
- Intellectual disability

Often, this group
of disorders will be
on your
differential
diagnosis list
together

Communication Disorders – Diagnostic Features

Language
Disorder

Speech Sound
Disorder

Child-Onset
Fluency Disorder

Social (Pragmatic)
Communication
Disorder

Language disorder – persistent difficulties in comprehension and/or production of language

- Results in functional limitations
- Presents as reduced vocabulary, limited sentence structure and limited conversation
- Deficits in spoken and/or written communication or sign language.
- Can involve expressive and/or receptive abilities.

Communication Disorders – Diagnostic Features

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Speech sound disorder: persistent difficulty with speech and sound production that interferes with speech intelligibility or prevents verbal communication

- Articulation of speech sound is impaired to the point that it interferes with ability to communicate

*Speech sound should be 100% intelligible by 4 years of age

Communication Disorders – Diagnostic Features

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Child-onset fluency disorder: stuttering

- Clinically significant impairment in speech fluency and timing

Communication Disorders – Diagnostic Features

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Social pragmatic communication disorder: impairment in the use of language for social exchanges, ability to adapt communication style to context of interaction, ability to follow conventional and cultural norms for conversation, ability to understand ambiguous language.

- Differentiate from autism

Speech & Language Disorders: Work-up and Management

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Autism Spectrum Disorder

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A lifelong neurodevelopmental disorder associated with deficits in social communication and social interaction and accompanied by restrictive/repetitive behaviors.

ASD:

Diagnostic Features

Deficits in social communication and social interaction

Social-emotional reciprocity

Nonverbal communication behaviors

Developing, maintaining, understanding relationships

Need at least 1 from each social category

Restrictive, repetitive behaviors, interests or activities

Stereotyped, repetitive movements or speech

Insistence on sameness, inflexible adherence to routines, ritualized behavior

Restricted interests

Hyper or hypo-reactivity to sensory input

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Cognitive development:
most individuals with ASD
have intellectual abilities in
the normal range (IQ > 70)

Need at least 1 from 2 of 4 non-social categories

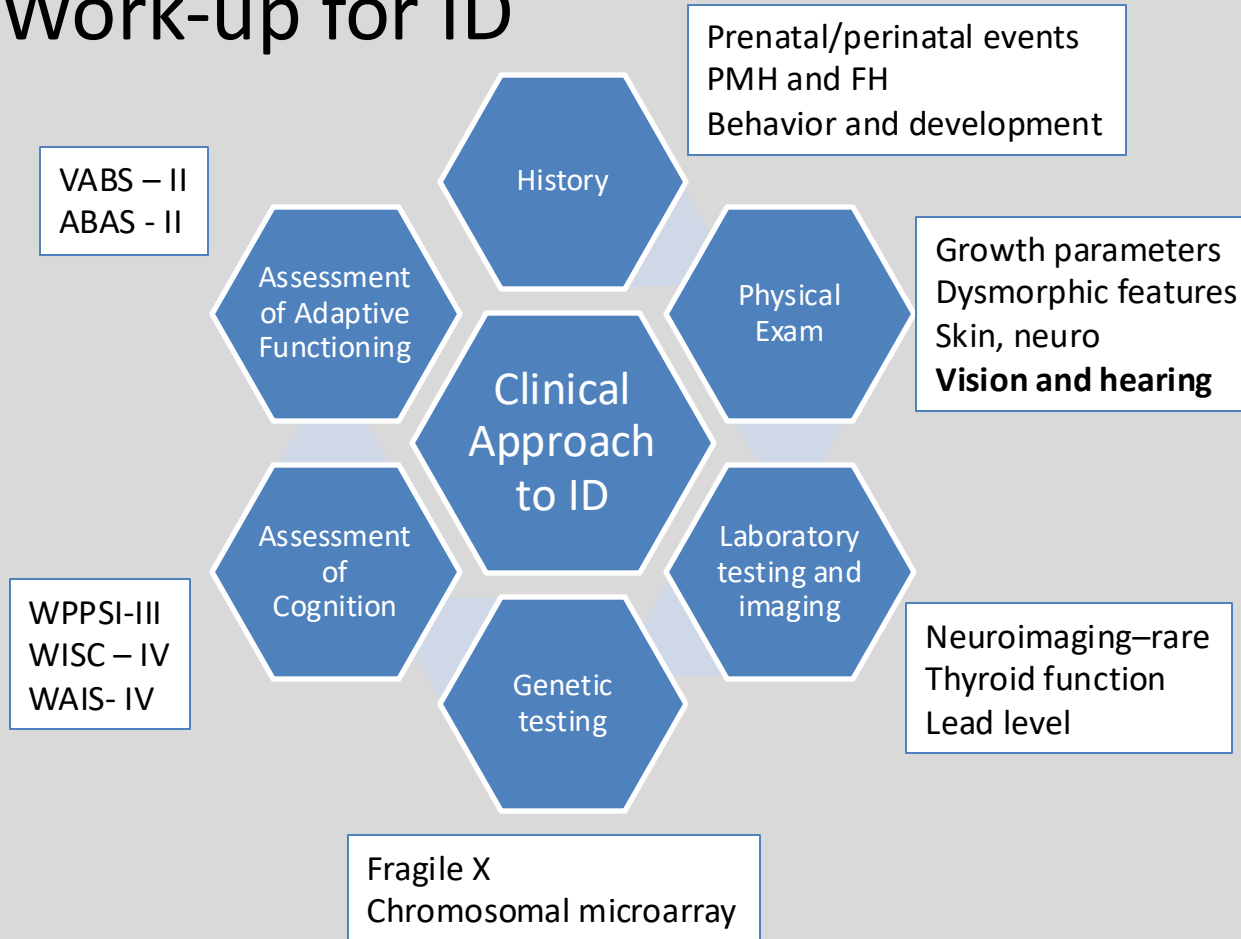
Diagnosing ASD

- Early diagnosis and intervention is optimal
 - History
 - Assess hearing and vision
 - Consider comorbid conditions
- Screening tools used by PCP
 - Infant-Child Checklist: < 18 months
 - **M-CHAT-R**: 18 – 30 months
 - Child Autism Screening test: preschool/school age
- Refer for in depth evaluation by Developmental Pediatrician and multidisciplinary team
 - **Do not delay management steps while awaiting this evaluation and formal diagnosis**
 - Refer to appropriate services **ASAP**

Management of ASD

- Educational and behavioral interventions
 - Applied Behavioral Analysis (ABA)
 - Speech therapies
 - Functional communication
 - Occupational therapies
 - Social skill support
- Medications for specific symptoms – inattention, anxiety, etc.
- Ongoing monitoring for comorbidities
- Support for family

Work-up for ID



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Motor Disorders

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Developmental Coordination Disorder

- Coordinated motor skills
- Clumsiness, slowness, inaccuracy

Stereotypical Movement Disorder

- Repetitive, purposeless motor behaviors
- May or may not be self-injurious

Tourette Syndrome

- Multiple motor AND one or more vocal tics
- Present for at least one year

Persistent/Chronic Tic Disorder

- Motor OR vocal tics
- Present for at least one year

Return to Patient 1

- Developmental screening with Ages & Stages demonstrated a delay in communication
- M-CHAT-R was not concerning for ASD
- Referred for audiology testing and formal speech and language evaluation
 - Demonstrated bilateral hearing loss
 - Speech and language evaluation confirmed delayed speech
 - Sent to ENT for further evaluation and hearing aids
 - Speech therapy started
 - At his 30 month well child visit he demonstrated improvement

Patient 2

- 7 year old male in second grade brought to you specifically because of school problems
 - Poor grades – C, D, F
 - Teacher recommends he repeat
 - For last several months: getting in trouble at school for talking back, getting out of seat, not listening, fighting with peers
 - Birth Hx: Prematurity
 - PMH: required speech therapy for speech delay
 - Fam Hx: ADHD, learning problems
 - Social Hx: several stressors

Differential Diagnosis for Patient 2

- Learning disorder
- Behavior problem
 - ODD
 - Conduct disorder
- ADHD
- Intellectual disability
- Mood disorder
- Reaction to psychosocial stressors

Differential
diagnosis for
school
dysfunction

Learning Disorder

Discrepancy between intellectual ability
and academic performance

NOT due to:

- Vision or hearing problems
- Motor disability
- Environmental, cultural or economic disadvantage
- Intellectual disability

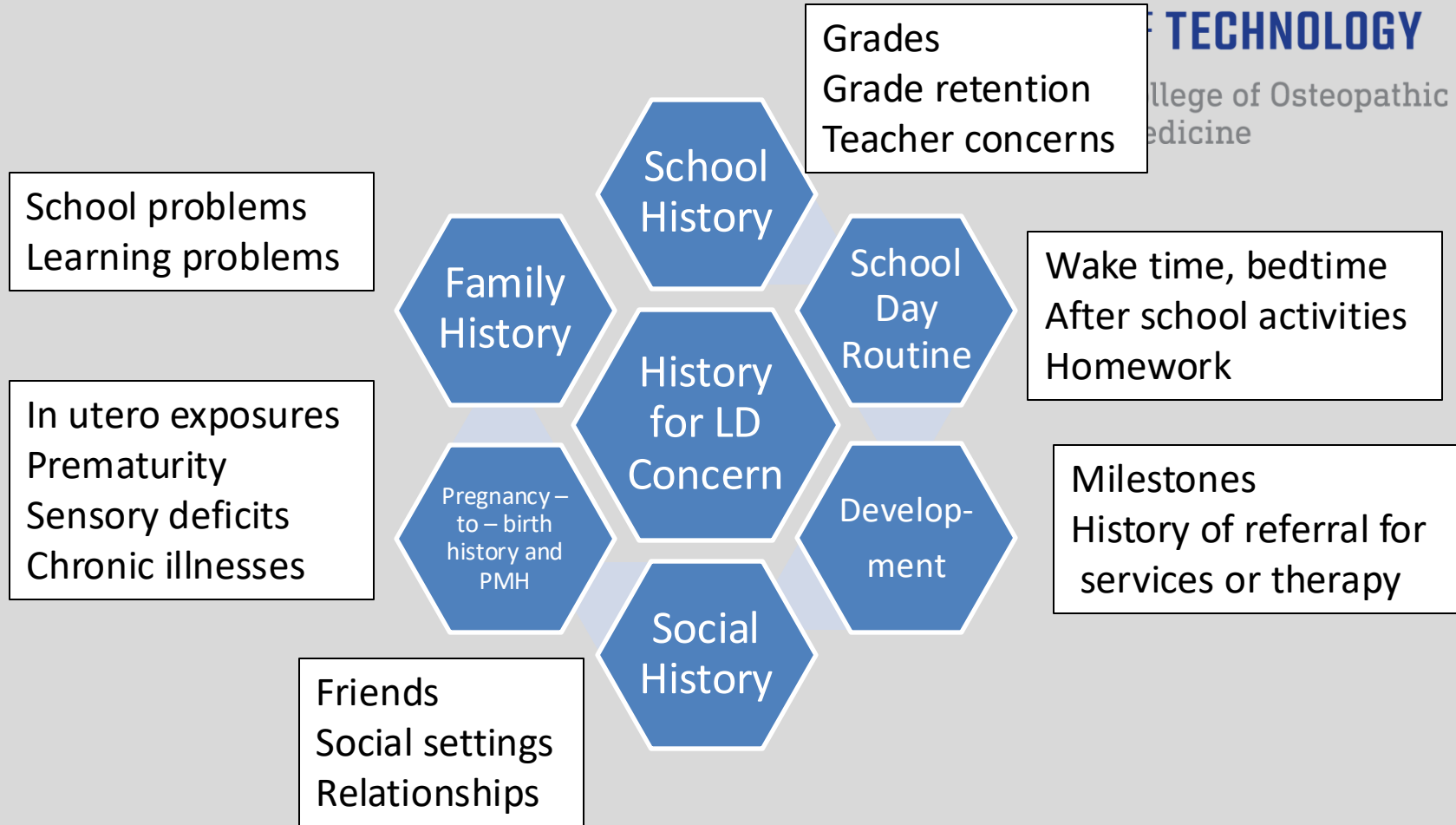
Risk factors:

- Prematurity
- Positive family history
- Preschool developmental problems
 - Especially speech and language: *high association with reading LD

History is Key to Diagnosis

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Ongoing Management Can Be Complex

- Advocate for family
 - Ensure proper services received
- Provide a medical home
 - Ongoing assessment for functional or mental health problems
 - Review IEP (Individualized Education Program), school accommodations, therapies, report cards
 - Ongoing behavioral and developmental surveillance

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Summary Slide

- Normal child development is variable
- Delays can occur in any developmental domain
- It is important to monitor closely for developmental delay and neurodevelopmental disorders because intervening early is key
- Always consider a broad differential diagnosis and comorbid conditions
- A thorough history is key
- Education and support for parents is very important